Quantitative size transfer at arbitrary fixed depth

A conditional-tuple proof for normalized Invariant Graphon Networks
Research notebook contribution · 2 November 2026

Status: AI-assisted mathematical draft; needs independent review. Novelty is unassessed. The full website problem remains open. Finite computations support examples and do not certify the general proof.

1. Main draft theorem

Fix a finite-depth scalar-output network built from normalized partial-matching tensor operators, channel mixing, constant biases and Lipschitz pointwise activations. Tensor orders, widths, weights and depth are independent of sample size. Section 2 defines the exact class, including which diagonal operations it omits.

Let N_n be its output on a simple loopless Bernoulli graph sampled from any symmetric measurable graphon W, with bounded measurable signal f. There is a deterministic sampling limit $\mu = \mu_N(W,f)$. If K is the largest tensor order and $n,m \geq 2K$, then

$$E|N_n - \mu| \leq P_L/\sqrt{(n-K)} + Q_L/n,$$

$$E|N_n - N_m| \leq P_L[(n-K)^{-1/2}+(m-K)^{-1/2}] + Q_L(1/n+1/m).$$

The constants below are explicit in the architecture, weights, activation constants and signal bound, and uniform over W and f in that bounded class. The second inequality follows by the triangle inequality; in particular it holds for independent samples. No polynomial expansion or graphon smoothness is used.

What advances beyond the initial notebook

The previous local-kernel bound stopped before a nonlinear activation after pooling. The proof here replaces each entire intermediate tensor by a bounded function of a tuple's latent positions and internal edge bits. Conditional averaging then carries the estimate through every fixed layer. Section 8 gives a genuinely nonlocal three-layer ReLU example with a matching root-n obstruction.

Section 6 also gives an explicit recursion for continuous activation moduli and the worst-case sharp Hölder exponent $\frac{1}{2}\prod\alpha_\ell$. Neither claim supplies one universal rate for all continuous activations.

2. Exact model and constants

Draw $X_1,\dots,X_n$ independently uniformly on $[0,1]$. Independently draw one uniform U_{ij} for each unordered pair $i<j$, put $A_{ij}=A_{ji}=1\{U_{ij}\le W(X_i,X_j)\}$, and set $A_{ii}=0$. Assume $|f|\le R$. The order-two input has channels $H_{0,1}(i,j)=A_{ij}$ and $H_{0,2}(i,j)=f(X_i)$.

A basis map γ from order p to order q is a bijection between a subset of s input positions and a subset of s output positions. The other $r_\gamma=p-s$ input positions are free. For each output tuple $i\in[n]^q$,

$$(T_{\gamma,n}H)(i) = n^{-r_\gamma}\,\Sigma_{j\in[n]^{r_\gamma}}\,H(v(i,j)).$$

Here matched input positions copy their matched output indices, and free positions take j in a fixed order. All free index tuples, including repetitions, are summed. Orders zero are allowed. These maps average, permute and replicate; they do not extract within-input diagonals or multiply by output equality masks. The theorem is for this partial-matching class, not the full partition basis of arbitrary finite IGNs.

$$H_{\ell,a} = \sigma_\ell(b_{\ell,a} + \Sigma_{b,\gamma} w_{\ell,a,b,\gamma} T_{\gamma,n} H_{\ell-1,b}),$$
$$k_0=2,\ k_L=0,\ N_n=H_{L,1}.$$

Take the last width to be one; its activation may be the identity. Write $K=\max k_\ell$, $a_\ell=\max_a \Sigma_{b,\gamma}|w_{\ell,a,b,\gamma}|$ and $b_\ell=\max_a|b_{\ell,a}|$. Let r_ℓ be the largest free order of a nonzero-weight map (zero if none), and $d_\ell=k_\ell r_\ell+r_\ell(r_\ell-1)/2$. Let L_ℓ be a Lipschitz constant on the reachable preactivation interval.

Every T has sup norm at most one. Thus $M_0=\max(1,R)$, and any uniform bounds M_ℓ for the finite tensors and the limiting fields may be used. One valid choice is

$$M_\ell = |\sigma_\ell(0)| + L_\ell(b_\ell+a_\ell M_{\ell-1}).$$

With $P_0=Q_0=0$, the theorem's constants are

$$P_\ell = L_\ell a_\ell(P_{\ell-1}+M_{\ell-1}r_\ell),$$
$$Q_\ell = L_\ell a_\ell(Q_{\ell-1}+2M_{\ell-1}d_\ell).$$

For continuous activations use $M\ell=\sup\{|\sigma\ell(z)|: |z|\le b\ell+a\ell M\ell-1\}$. Boundedness then follows from compactness at each fixed layer. Constants can be coarse and may grow rapidly with depth.

3. The conditional-tuple sampling limit

For an injective ordered k-tuple I, its type τ_I consists of the latent positions $(X_i)_{i\in I}$ and every internal Bernoulli edge bit between these vertices. Signals are f at these positions. Denote the space of such abstract types by Types_k; Types_0 is a singleton. Fields below are bounded measurable functions on these spaces.

The limiting partial-matching operator

Given an output type $\tau\in\mathrm{Types}_q$, retain the s matched latent positions and their mutual edge bits. Draw r_γ fresh independent uniform latent positions. Draw all input edges involving at least one fresh position as conditionally independent Bernoulli variables with probabilities W. Form the resulting p-type, in the input order prescribed by γ. For a bounded input field φ define

$$(\mathbb{P}_\gamma\varphi)(\tau) = \mathrm{E}[\varphi(\text{fresh input type}) \mid \tau].$$

Only the matched output vertices occur in the input type. Conditioning on the full output type, including unused output vertices, causes no difficulty. The fresh latent positions and required edge uniforms are independent of that conditioning. This explicit construction defines $\mathbb{P}$ even on output types having probability zero.

Initialize $\varphi_{0,1}(x_1,x_2,e_{12})=e_{12}$ and $\varphi_{0,2}=f(x_1)$. Recursively set

$$\varphi_{\ell,a}(\tau) = \sigma_\ell\left(b_{\ell,a} + \Sigma_{b,\gamma} w_{\ell,a,b,\gamma}\,\mathbb{P}_\gamma\varphi_{\ell-1,b}(\tau)\right).$$

Integration preserves bounded measurability, and $|\mathbb{P}_\gamma\varphi|\le||\varphi||\infty$. Induction gives $|\varphi_{\ell,a}|\le M_\ell$. At order zero the type is empty, so $\varphi_{L,1}$ is the deterministic number μ in the theorem.

Why retaining Bernoulli bits matters

The limit need not be the original network evaluated directly on the weighted graphon W. A nonlinear local expression A_{ij}^2 equals A_{ij} on a simple graph. Its normalized mean tends to $\int W$, whereas the corresponding weighted expression gives $\int W^2$. A ReLU realization is $\rho(2A_{ij}-1)$. This distinction survives at large n.

The present result compares sizes sampled from the same (W,f) through their common Bernoulli sampling limit. It does not establish cut-distance continuity of the original weighted-input network, transfer between different graphons, or trained-model generalization.

4. Conditional averaging lemma and proof

Lemma. Fix a partial matching from p to q with r free input positions, $n \geq p+q$, and $M \geq 0$. Let H_n be any bounded, possibly globally data-dependent input tensor, and φ a local p-type field, both bounded in absolute value by M. Suppose for every injective input tuple I, $|H_n(I) - \varphi(\tau_I)| \leq e$. Then for every injective output tuple B,

$$|T_{\gamma,n} H_n(B) - \mathbb{P}_\gamma \varphi(\tau_B)| \leq e + Mr/\sqrt{(n-q)} + (2M/n)[qr + r(r-1)/2].$$

1. Remove collisions explicitly

Call a free tuple good if its r indices are distinct and avoid all q output vertices. Good tuples have mass $g = (n-q)_r / n^r$, where $(a)_r$ is the falling factorial. Under independent uniform free indices, a union bound gives

$$1 - g \leq [qr + r(r-1)/2]/n.$$

For good tuples the assembled input is injective. Replacing H_n by φ in their normalized average costs at most $ge \leq e$ in unconditional expected absolute error. This step uses no independence of H_n entries. The contribution from bad tuples has absolute value at most $M(1-g)$.

2. Concentrate only the local field

Let U be the average of φ over good ordered free tuples, normalized to total mass one. Conditional on τ_B, each summand has mean $h = \mathbb{P}_\gamma \varphi(\tau_B)$. Two summands using disjoint fresh vertex sets are conditionally independent: shared output latents and internal edges have been fixed, and the fresh latents and edge uniforms actually used by the two kernels are disjoint. Unused cross-edges play no role.

For two independent uniformly chosen good free tuples, the probability their fresh sets overlap is at most $r^2/(n-q)$: each of the r^2 pairs of positions agrees with probability $1/(n-q)$. Every remaining conditional covariance has absolute value at most M^2, by Cauchy–Schwarz and the bounded range. Consequently

$$\mathrm{Var}(U \mid \tau_B) \leq M^2 r^2/(n-q),$$
$$E|U-h| \leq Mr/\sqrt{(n-q)}.$$

For $r=0$ interpret $U=h$ and the fluctuation term as zero, including the boundary $n=q$; there are no collisions. For $r>0$, $n \geq p+q$ guarantees that good tuples exist. Finally compare the good part gU plus the bad part to h: its error is bounded by $g|U-h| + 2M(1-g)$. Adding the earlier replacement error proves the lemma.

Dependence audit: applying conditional concentration directly to the actual hidden tensor would be unjustified: it depends on the entire graph. The unconditional H-to-φ comparison precedes the conditional variance calculation, which is applied only to the local field φ.

5. Induction and a concentration corollary

For each layer, let $e_\ell(n)$ be the maximum over channels and injective output tuples of $E|H_{\ell,a}(I)-\varphi_{\ell,a}(\tau_I)|$. Exchangeability makes the tuple choice immaterial, but a maximum suffices. Initially $e_0=0$. Since $n\ge 2K$, the lemma applies to every basis map. Its q is $k_\ell\le K$, while $r\le r_\ell$.

Apply the triangle inequality to channel mixing and then the Lipschitz activation. Both preactivations lie in the same bounded interval. Monotonicity of $qr+r(r-1)/2$ in r gives

$$e_\ell(n) \le L_\ell a_\ell [e_{\ell-1}(n) + M_{\ell-1} r_\ell/\sqrt{(n-K)} + 2M_{\ell-1} d_\ell/n].$$

Insert the definitions of P_ℓ and Q_ℓ from Section 2 to obtain $e_\ell(n)\le P_\ell/\sqrt{(n-K)}+Q_\ell/n$ by induction. At order zero this is precisely the first theorem bound. Comparing both scalar outputs to μ proves the second bound. No step approximates σ by a polynomial.

High-probability form for Lipschitz layers

Let $\Lambda=\prod_{\ell=1}^{L} L_\ell a_\ell$. Each T is a contraction in normalized entrywise L^1: average $|TH|$ over all output tuples, use the triangle inequality, and note that matched output indices and free indices together are independent uniform input coordinates. Replicated unused output positions cancel. Thus the network is Λ-Lipschitz in the maximum-channel normalized L^1 input distance.

Resampling one latent X_i changes the adjacency only in one row and column and the signal only in one first-coordinate slice. Its output effect is at most $2M_0\Lambda/n$. Resampling one unordered edge uniform changes two entries, giving effect at most $2\Lambda/n^2$. These X and U variables are independent. Their squared bounded differences sum to at most $\Lambda^2(4M_0^2+2)/n$. McDiarmid’s inequality therefore gives, for $\Lambda>0$,

$$\Pr(|N_n-EN_n|\ge t) \le 2\exp\{-nt^2/[\Lambda^2(2M_0^2+1)]\}.$$

Combining with $|EN_n-\mu|\le E|N_n-\mu|$ yields, with probability at least $1-\delta$,

$$|N_n-\mu| \le P_L/\sqrt{(n-K)}+Q_L/n + \Lambda\sqrt{[(2M_0^2+1)\log(2/\delta)/n]}.$$

If $\Lambda=0$, some layer is constant with respect to the preceding tensors, so the final output is deterministic and sample-size independent. The probability statement is then interpreted directly, without division by zero.

6. Continuous moduli and Hölder rates

For continuous σ_ℓ, put $Z_\ell = b_\ell + a_\ell M_{\ell-1}$. Let $\omega_\ell(t)$ be its modulus of continuity on $[-Z_\ell, Z_\ell]$, constant beyond that interval's diameter. Define

$$\Omega_\ell(t) = \inf_{\delta>0} (1 + t/\delta)\omega_\ell(\delta),\ t>0;\ \Omega_\ell(0)=0.$$

Subdividing a segment into steps of length at most δ shows $\omega_\ell(t) \le (1+t/\delta)\omega_\ell(\delta)$. Hence Ω majorizes ω. It is nondecreasing and concave as an infimum of affine functions of t. Taking $\delta = t$ gives $\Omega(t) \le 2\omega(t)$, so it tends to zero at zero, and the extension there remains concave. Jensen's inequality therefore applies to Ω.

Write $\eta_\ell(n) = M_{\ell-1}[r_\ell/\sqrt{(n-K)} + 2d_\ell/n]$. With $u_0(n) = 0$ define

$$u_\ell(n) = \Omega_\ell(a_\ell[u_{\ell-1}(n) + \eta_\ell(n)]).$$

The lemma bounds the mean preactivation discrepancy by $a_\ell(e_{\ell-1} + \eta_\ell)$. The pointwise discrepancy is controlled by ω, then Ω; Jensen and monotonicity give $e_\ell \le u_\ell$ inductively. Thus $E|N_n - N_m| \le u_L(n) + u_L(m)$. Every fixed finite-depth network converges, with an explicit rate once its activation moduli are supplied.

Closed form for Hölder activations

Suppose $|\sigma_\ell(x) - \sigma_\ell(y)| \le H_\ell |x-y|^{\alpha_\ell}$ on the reachable interval, with $0 < \alpha_\ell \le 1$. Set $\beta_0 = 1/2$ and $C_0 = 0$. For $n \ge 2K$ define constants

$$\beta_\ell = \alpha_\ell \beta_{\ell-1},\ T_\ell = \sqrt{2}\, r_\ell + 2d_\ell,$$
$$C_\ell = H_\ell \{a_\ell [C_{\ell-1} + M_{\ell-1} T_\ell]\}^{\alpha_\ell}.$$

Then $E|N_n - \mu| \le C_L n^{-\beta_L}$, where $\beta_L = \tfrac{1}{2}\prod \alpha_\ell$. Indeed $\eta_\ell \le M_{\ell-1} T_\ell n^{-1/2}$, $\beta_{\ell-1} \le 1/2$, and Jensen for t^{α_ℓ} gives the displayed recursion. Setting all $\alpha_\ell = 1$ recovers a root-n power; the earlier P,Q separate its lower-order collision correction.

7. The Hölder exponent is sharp in this class

This is a worst-case statement over networks in the specified class, not a claim about every fixed architecture. It also separates activation regularity from graphon smoothness.

Take $W \equiv 0$ and $f = +1$ on half of $[0,1]$, -1 on the other half. A normalized integration of the replicated signal gives the sample mean $S_n = n^{-1}\Sigma_i f(X_i)$. Use scalar-output activations $\sigma_\ell(t) = |t|^{\alpha_\ell}$ successively; between them use identity linear maps. Each activation is globally α_ℓ-Hölder with constant one. If needed, an identity final readout has exponent one.

$$N_n = |S_n|^A,\ A = \prod \alpha_\ell,\ \mu = 0.$$

For independent equal-size samples S_n and S'_n, the two-dimensional central limit theorem gives $(\sqrt{n}\, S_n, \sqrt{n}\, S'_n)$ converging in distribution to independent standard normal variables Z, Z'. The transformed differences are uniformly integrable: $0 < A \leq 1$ and the second moments of the normalized sample means are one, so a power greater than one of the differences has a uniform bound. Therefore

$$n^{A/2} E\big||S_n|^A - |S'_n|^A\big| \to E\big||Z|^A - |Z'|^A\big| > 0.$$

The limiting expectation is finite and positive because $|Z|^A$ is nonconstant. This proves that the exponent $A/2$ in Section 6 cannot be improved uniformly over this network class. In particular no $o(n^{-1/2})$ equal-size transfer rate holds uniformly for Lipschitz activations.

What this lower bound does and does not use

This construction uses signal fluctuations, not edge randomness. Section 9 supplies a separate three-layer example whose root-n bias comes entirely from graph edges and a nonlinear threshold after common-neighbor pooling. Both are fixed networks: parameters do not change with n.

If an activation has only continuity and no prescribed modulus, Section 6 gives a modulus-dependent recursion, but no common power follows. This is consistent with the earlier notebook construction of arbitrarily slow continuous networks. The new bound refines that observation by identifying the sharp power when Hölder regularity is specified.

8. A genuinely nonlocal three-layer network

Let $\rho(t)=\max(t,0)$, fix $\theta\in[0,1]$, and define the order sequence $2\to3\to2\to0$:

$$H_1(i,j,k)=\rho(A_{ik}+A_{jk}-1)=A_{ik}A_{jk},$$
$$C_{ij}=n^{-1}\Sigma_k H_1(i,j,k),$$
$$H_2(i,j)=\rho(C_{ij}-\theta),$$
$$N_n=n^{-2}\Sigma_{i,j}H_2(i,j).$$

This has three affine layers, two ReLUs, and an identity readout. The middle activation acts after a global sum over k, so it lies outside the previous fixed local-kernel argument. Every operator is a partial matching with the required normalization, including repeated tuples and $A_{ii}=0$.

$$q(x,y)=\int_0^1 W(x,z)W(y,z)dz,$$
$$\mu_\theta=\int_{[0,1]^2}(q(x,y)-\theta)_+\,dxdy.$$

A sharper direct bound: for every W and $n\geq2$,

$$E|N_n-\mu_\theta| \leq (3/2-\theta)/\sqrt{n} + (3-\theta)/n.$$

Proof with all finite-size terms

For $i\neq j$, condition on X_i,X_j. The $n-2$ terms $A_{ik}A_{jk}$, $k\notin\{i,j\}$, are independent Bernoulli$(q(X_i,X_j))$; terms $k=i,j$ vanish. The conditional standard deviation of C_{ij} is at most $\sqrt{(n-2)/(2n)}\leq1/(2\sqrt{n})$, and its conditional mean differs from q by at most $2/n$. Since ρ is 1-Lipschitz, replacing every off-diagonal H_2 by $h(X_i,X_j)=(q-\theta)_+$ costs at most $1/(2\sqrt{n})+2/n$ in expected normalized error.

Write $B=1-\theta$. The injective ordered-pair average U of h is unbiased for μ_θ, with $0\leq h\leq B$. Disjoint pairs are independent. A pair of uniformly selected injective ordered pairs shares a vertex with probability at most $4/n$, and each covariance is at most $B^2/4$. Hence $\mathrm{Var}(U)\leq B^2/n$ and $E|U-\mu_\theta|\leq B/\sqrt{n}$.

The off-diagonal fraction is $(n-1)/n$; the diagonal contribution is $(1/n)D$ with $0\leq D\leq B$. Thus replacing the ideal full expression by μ costs at most $B/\sqrt{n}+B/n$. Adding the two estimates proves the bound. Applying it at n and m gives the corresponding transfer inequality.

9. Matching root-n bias from graph edges

Take a constant graphon $W \equiv p$ with $0<p<1$ and choose $\theta=p^2$. Then $q \equiv p^2$ and $\mu_\theta=0$. For $i \neq j$, nC_{ij} has distribution $\mathrm{Bin}(n-2,p^2)$; for $i=j$, nC_{ii} is the degree, with distribution $\mathrm{Bin}(n-1,p)$. Therefore the exact expectation is

$$EN_n = [(n-1)/n^2]\, E[(\mathrm{Bin}(n-2,p^2)-n\theta)_+] + [1/n^2]\, E[(\mathrm{Bin}(n-1,p)-n\theta)_+].$$

The diagonal term is $O(1/n)$. For the first term, the centered binomial central limit theorem applies; the shift between $(n-2)p^2$ and $n\theta$ is constant and vanishes after division by $\sqrt{n}$. Uniformly bounded second moments of the normalized variables justify passing to expected positive parts. Consequently

$$\sqrt{n}\, EN_n \to \sqrt{[p^2(1-p^2)/(2\pi)]} > 0.$$

For independent sizes n and $m=n^2$, $E|N_n - N_{n^2}| \geq |EN_n - EN_{n^2}|$ is asymptotic from below to this positive constant divided by $\sqrt{n}$. Section 8 gives the matching upper order. Thus even this nonlinear common-neighbor network has a worst-case transfer obstruction of order $n^{-1/2}$.

Exact enumeration and asymptotic checks

With $p=1/2$ and $\theta=1/4$, the verifier enumerates all 33,866 labeled graphs at sizes 2 through 6 and compares the network's rational average against the independent binomial formula. They agree exactly:

n	graphs	exact E N
2	2	1/16
3	8	1/16
4	64	13/256
5	1024	91/1600
6	32768	1051/18432

The limiting constant is $\sqrt{(3/(32\pi))} \approx 0.172747075$. The computed $\sqrt{n}\, EN_n$ is 0.168314 at $n=64$, 0.172491 at $n=1024$ and 0.172716 at $n=8192$. Large-n sums are numerical, checked against exact rational binomial sums up to $n=128$. These checks corroborate the formulas; the asymptotic claim follows from the argument above.

10. A four-layer check and reproducibility

The accompanying verifier independently checks tensor sums, conditional-type integration, a nonlocal four-layer network and the preceding exact formulas. Its source, output and this report are linked together on the website.

Exact operator checks

For all 90 partial matchings with input and output orders in {0,1,2,3}, compare a vectorized tensor operator with explicit finite index sums. Separately construct the conditional-type operators on the equal-mass two-block graphon W=[[1/4,3/4],[3/4,1/2]]. All 3,544 conditional rows sum to one exactly using rational arithmetic. An explicit shared-vertex example has unconditional covariance 1/256 and zero covariance after conditioning on its shared latent block, illustrating why the conditional step matters.

A network with several nested nonlinear poolings

For that graphon put f=−1 on block 0 and +1 on block 1. Let T_{01} average the third input coordinate, T_0 average the second and third and replicate the second output position, $T_\varnothing$ average all inputs and replicate, and τ transpose the two outputs. Define

$$g_1(i,j,k)=\rho(A_{ik}+A_{jk}-1),$$
$$g_2(i,j,k)=\rho(\tfrac{1}{2}A_{ij}-\tfrac{1}{4}f(X_i)+1/10),$$
$$h_1=\rho(T_{01}g_1-1/4),$$
$$h_2=\rho((2/5)T_0g_1+(3/5)T_{01}g_2-1/10),$$
$$h_3=\rho((7/10)h_1+(1/5)\tau h_2+(1/10)T_\varnothing h_2-1/20),$$
$$N_n=n^{-2}\Sigma_{i,j}h_3(i,j).$$

Exact conditional-type integration gives μ=2603/32000=0.08134375. All intermediate channel amplitudes are at most one. Using r=(0,2,2,2), d=(0,5,5,1), a=(2,1,1,1) and all Lipschitz constants one gives K=3, P_4=6, Q_4=22. The resulting bound is valid but deliberately coarse.

n	mean absolute error	standard error
12	0.026476	0.001850
24	0.016217	0.000954
48	0.011228	0.000719
96	0.006231	0.000461

Monte Carlo: 120 independent samples per size, NumPy generator seed 112026. Standard errors describe this simulation, not a theorem confidence interval or an independent review. Exact Rademacher moment sums also check powers 1/2, 1/4 and 1/8. Run: python verify-deep-transfer.py (requires NumPy).

11. Literature position and next review

Primary sources checked on 2 November 2026

[1] Herbst and Jegelka, Higher-Order Graphon Neural Networks: Approximation and Cut Distance, ICLR 2025, arXiv version 3 (18 May 2025). Definition 4.3 and equations (9)–(12) motivate the no-diagonal partial-matching model. Theorem 5.2 / Corollary 5.3 give qualitative transfer; Theorem 5.4 / Appendix G.1 give a restricted analytic two-layer rate. Appendix G.4 discusses the difficulty of extending the expansion to deeper networks.

https://arxiv.org/html/2503.14338

[2] Levin, Ma, Díaz and Villar, On Transferring Transferability: Towards a Theory for Size Generalization, NeurIPS 2025. Appendix G.3 treats related graph networks and modified nonlinearities in a general transfer framework; its weighted-input continuity issues are distinct from direct convergence of simple Bernoulli samples. The sampling limit in Section 3 explicitly preserves edge bits.

NeurIPS 2025 primary conference paper, Appendix G.3

[3] Cai and Wang, Convergence of Invariant Graph Networks, 2022. This is related background on weighted inputs and sampling/denoising hypotheses; identifying an exact pre-existing theorem covering the present model remains part of the novelty audit.

https://arxiv.org/abs/2201.10129

What has been achieved locally

A self-contained draft proof now gives architecture-level root-n sampling bounds at arbitrary fixed Lipschitz depth in the specified partial-matching class, a modulus recursion, sharp worst-case Hölder powers, and an explicit nonlinear three-layer network with matching root-n bias. These extend the saved notebook beyond its local and polynomial arguments. The literature search is not exhaustive; publication novelty is not claimed.

What the next session must examine

Independent proof audit: check conditional independence with overlapping tuples, the unconditional replacement step, the collision correction, and every constant. Verify that the model exactly covers the intended IWN definition, without silently including within-input diagonal operations. An audit must either reproduce the proof independently or give a specific counterexample; additional Monte Carlo alone is insufficient.

Novelty and scope audit: search for the same arbitrary-depth sampling theorem under equivalent terminology and record attribution. Determine whether any omitted architectural operation changes the claim. Full partition-basis IGNs, unbounded/sparse inputs, parameters growing with n, optimal constants, and distribution shift are outside this theorem. These are limitations, not asserted counterexamples.

Provenance: continuation of revision 2026-11-01.1, task lipschitz-depth-three-rate, prior attempt initial-higher-order-transfer-rates. Author: AI-assisted research session, 2 November 2026. Review: needs-independent-review. No independent reviewer or formal proof assistant has certified these claims.